BDSD MPC Training Report

IDBS Reference: EXXXXX

Report Generated for CD19-CD20

Model Training Dataset: AR25-094

States in Model: IGG, VCC, Viability, Osmolality, Lactate

Inputs in Model: Normalized_Daily_Feed, Temperature_setpoint, pH_setpoint

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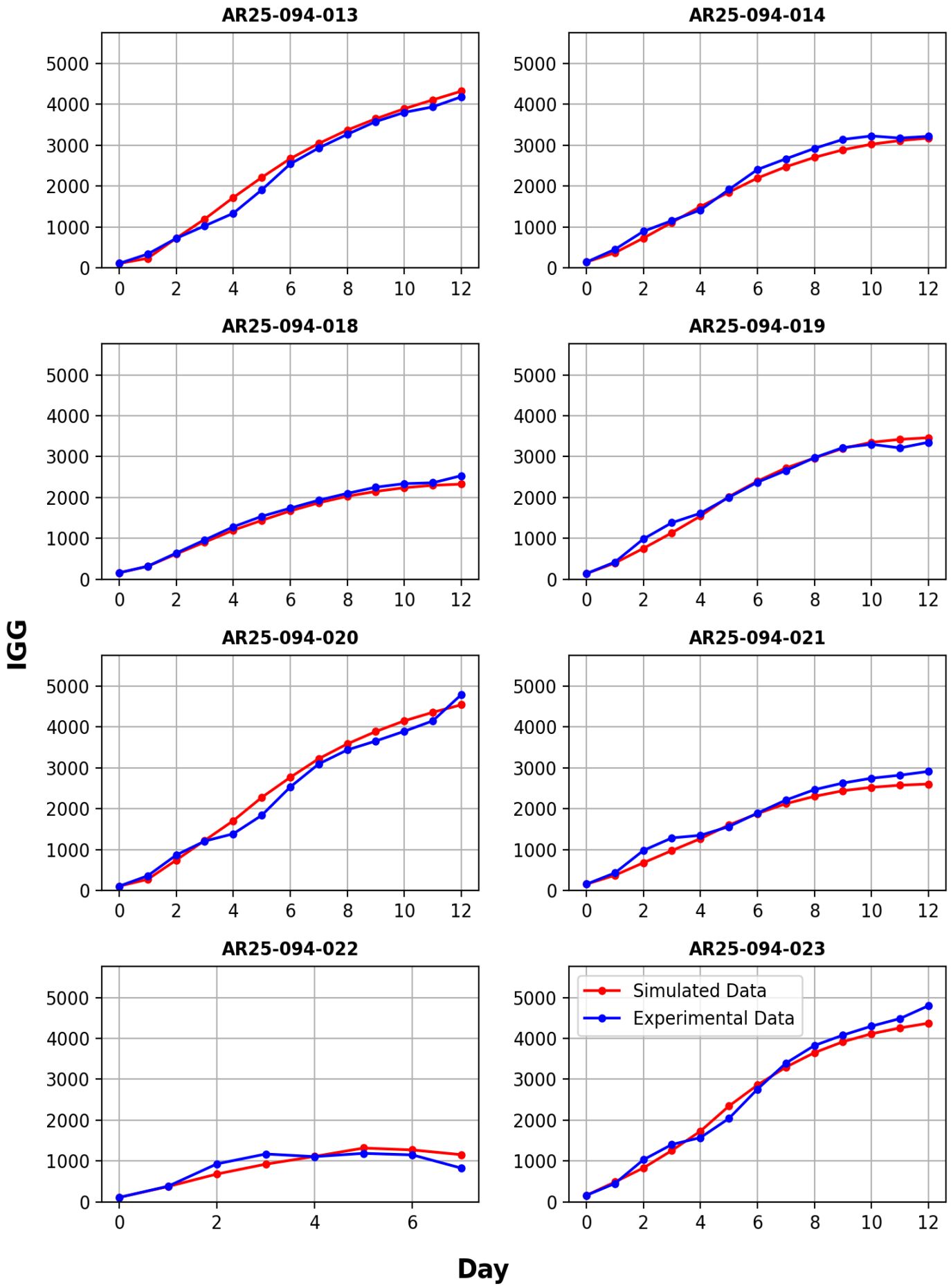
Report Date: 2025/11/12 16:10:17

Scaling Parameters

feature_names_in_	scale_	min_	data_min_	data_max_	data_range_
IGG	0.0	-0.022	104.998	4799.85	4694.852
VCC	0.036	-0.517	14.21	41.711	27.501
Viability	0.021	-1.039	50.6	99.3	48.7
Osmolality	0.007	-1.9	255.5	390.0	134.5
Lactate	1.058	-0.037	0.035	0.98	0.945
Normalized_Daily_Feed	38.137	0.0	0.0	0.026	0.026
Temperature_setpoint	1.0	-35.0	35.0	35.0	0.0
pH_setpoint	2.5	-17.25	6.9	7.3	0.4

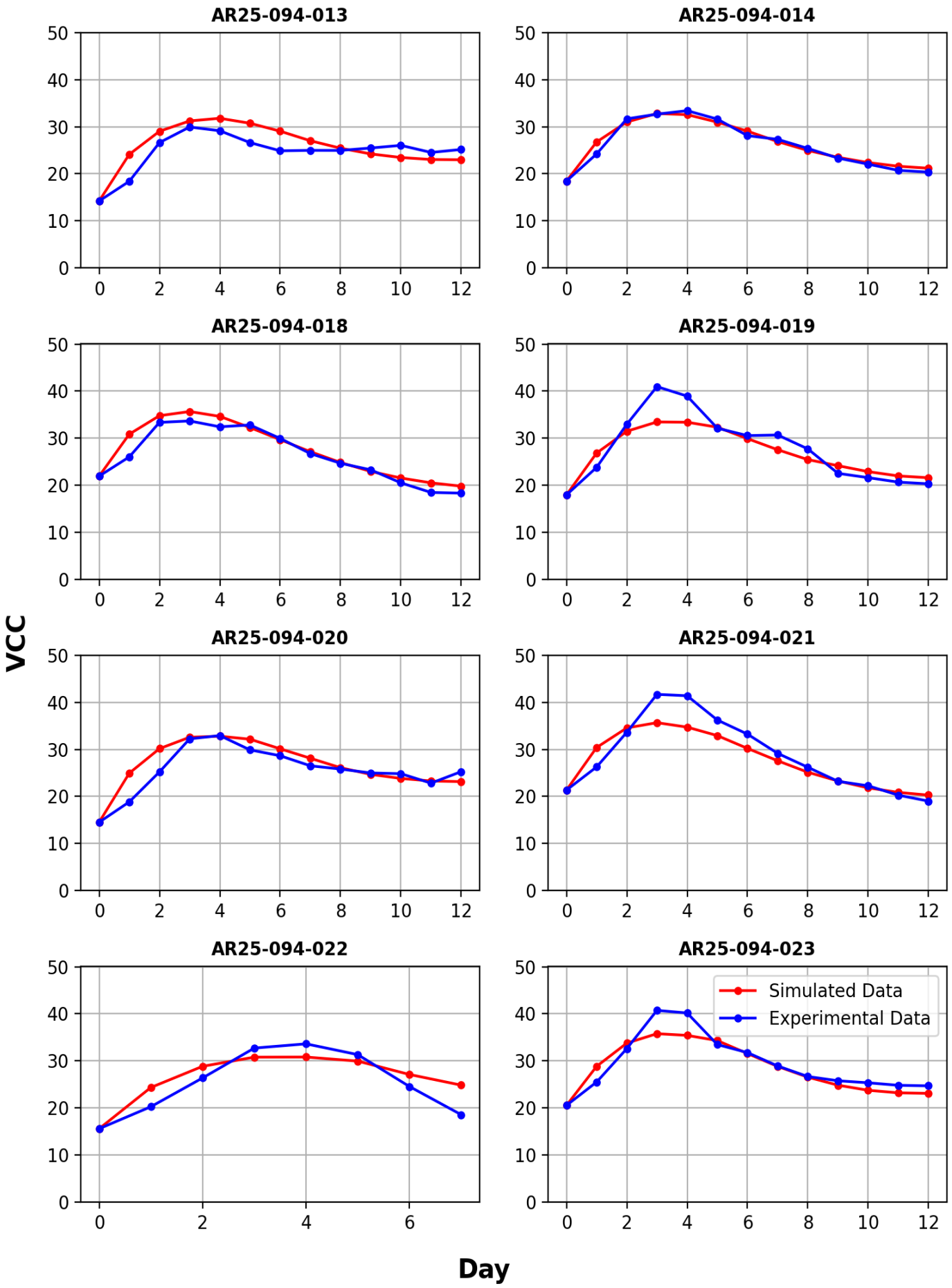
Model Training Dataset

State: IGG


AR25-094-013
AR25-094-014
AR25-094-018
AR25-094-019
AR25-094-020
AR25-094-021
AR25-094-022
AR25-094-023
IGG
Day
Simulated Data
Experimental Data

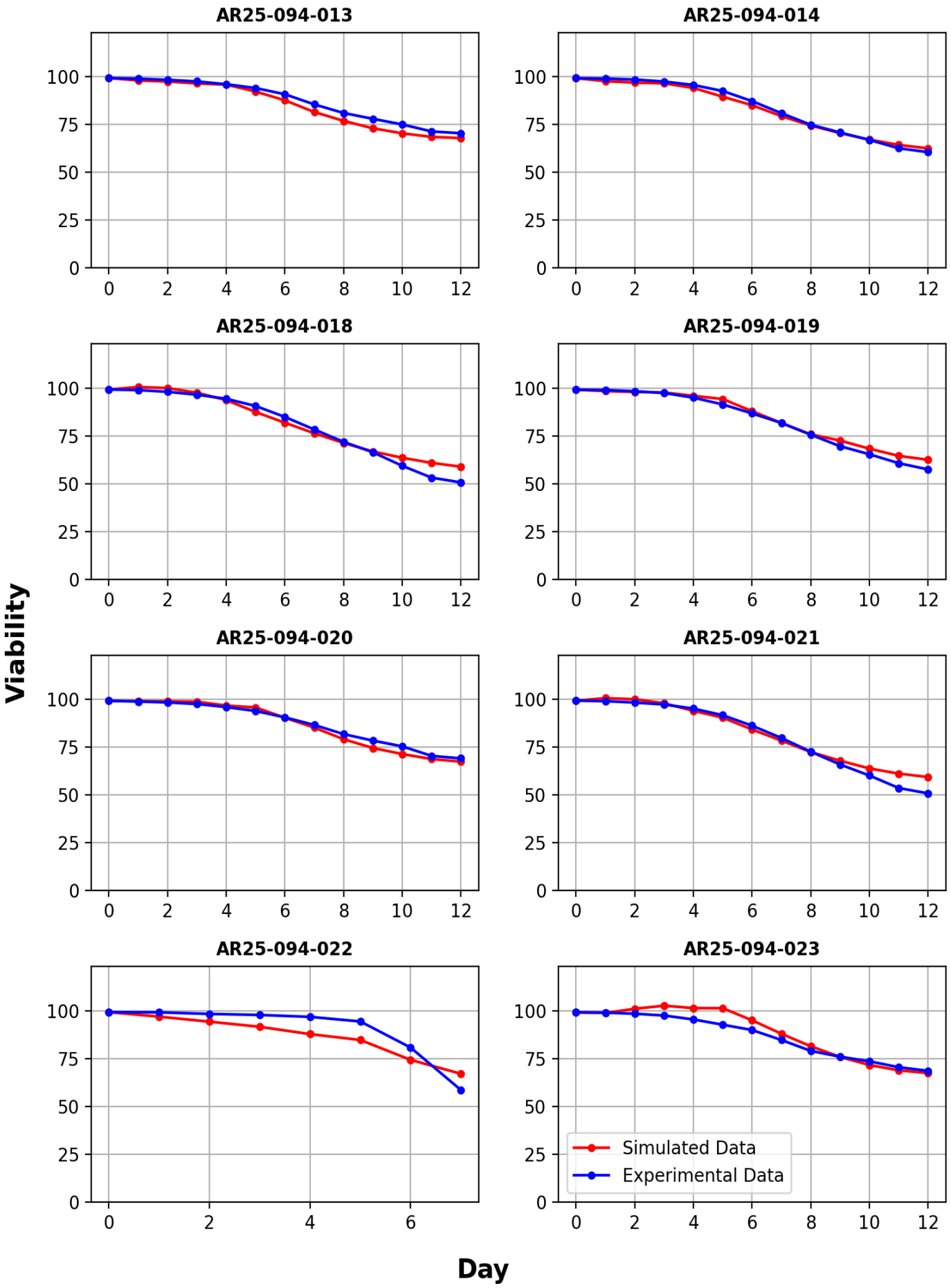
AR25-094-024
5000
4000
3000
2000
1000
0
0
2
4
6
8
10
12
IGG
Day

State: VCC


AR25-094-013
AR25-094-014
AR25-094-018
AR25-094-019
AR25-094-020
AR25-094-021
AR25-094-022
AR25-094-023
Simulated Data
Experimental Data
VCC
Day

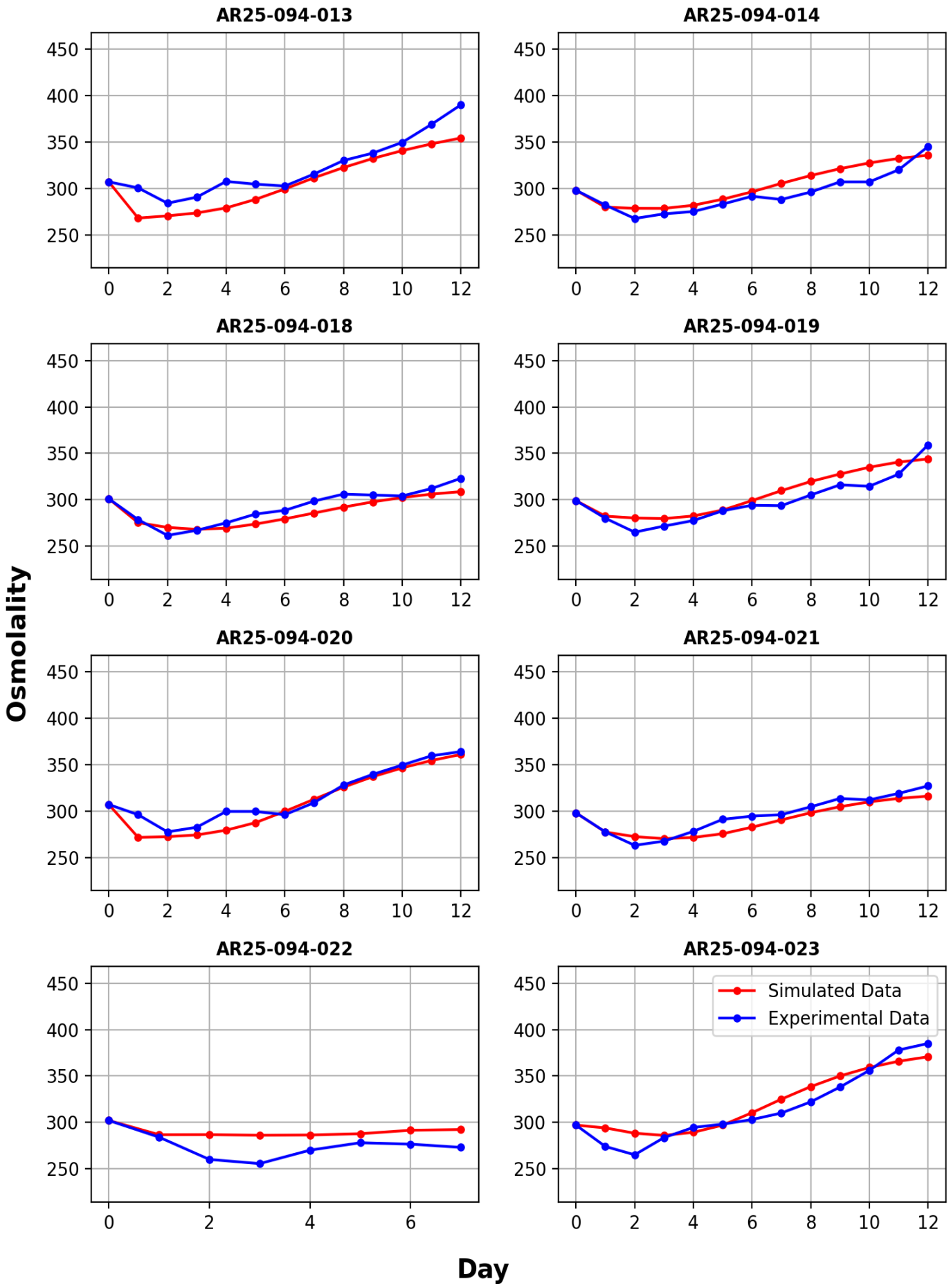
AR25-094-024
50
40
30
20
10
0
0
2
4
6
8
10
12
VCC
Day

State: Viability



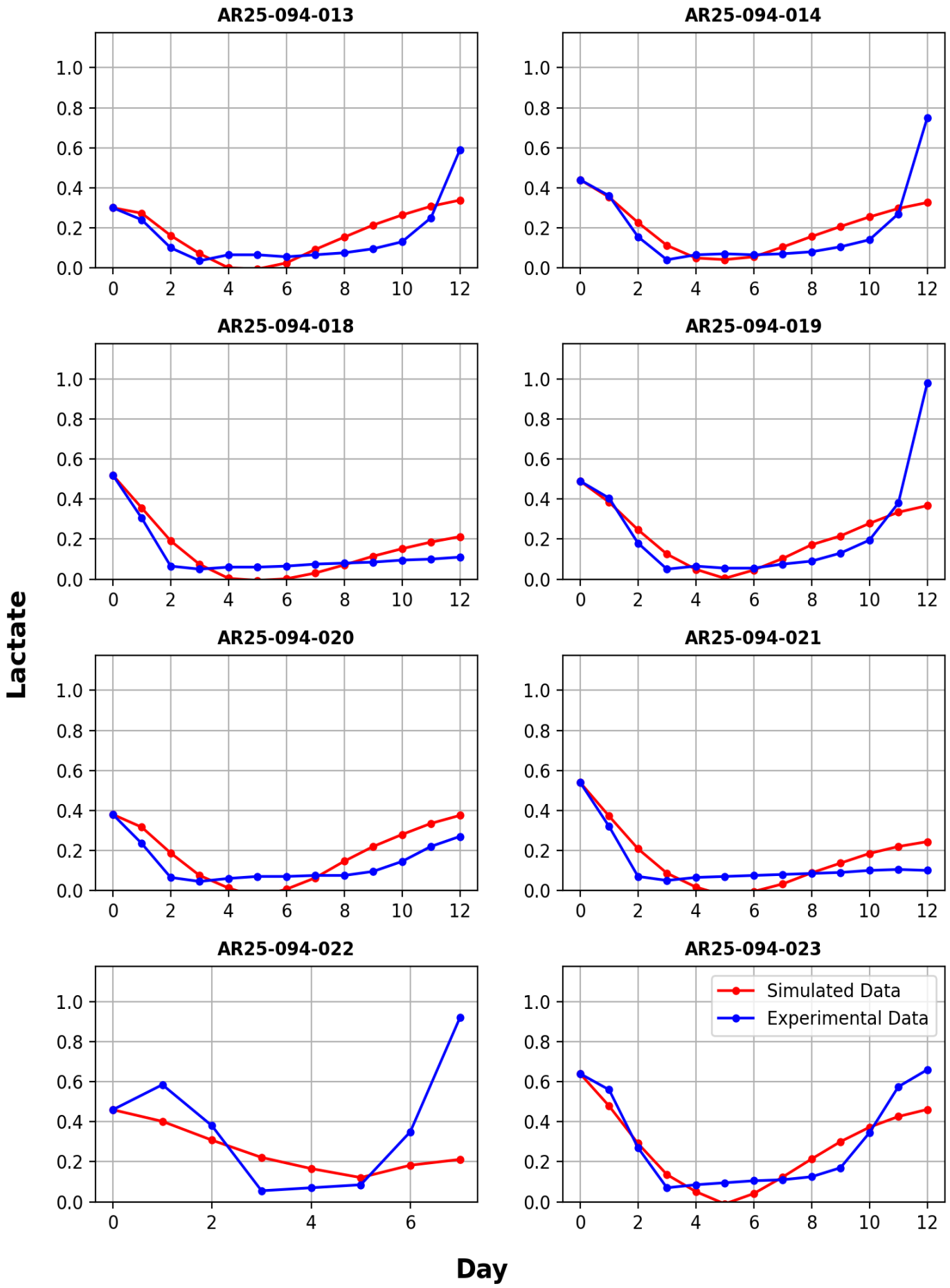
AR25-094-024
120
100
80
60
40
20
0
0
2
4
6
8
10
12
Viability
Day

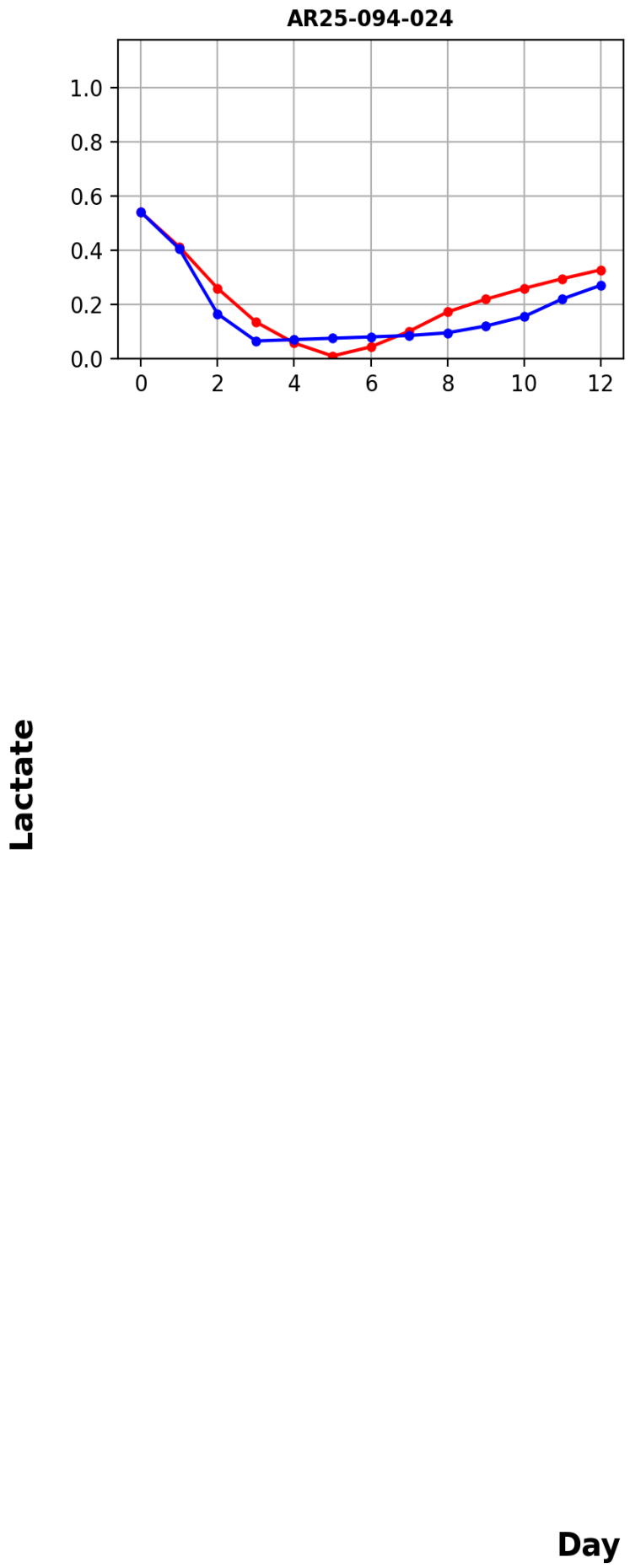
State: Osmolality



AR25-094-024
450
400
350
300
250
0
2
4
6
8
10
12
Osmolality
Day

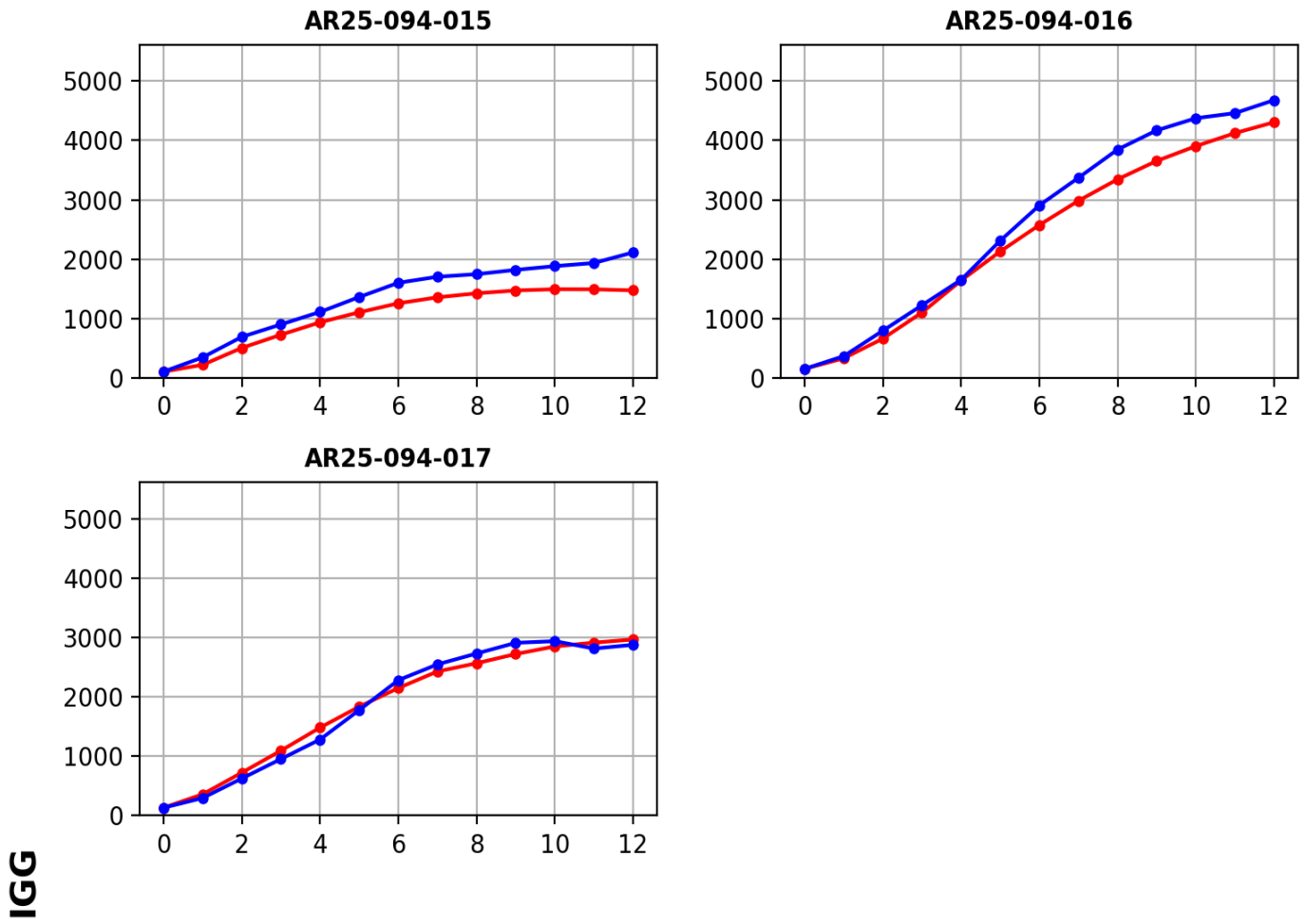
State: Lactate



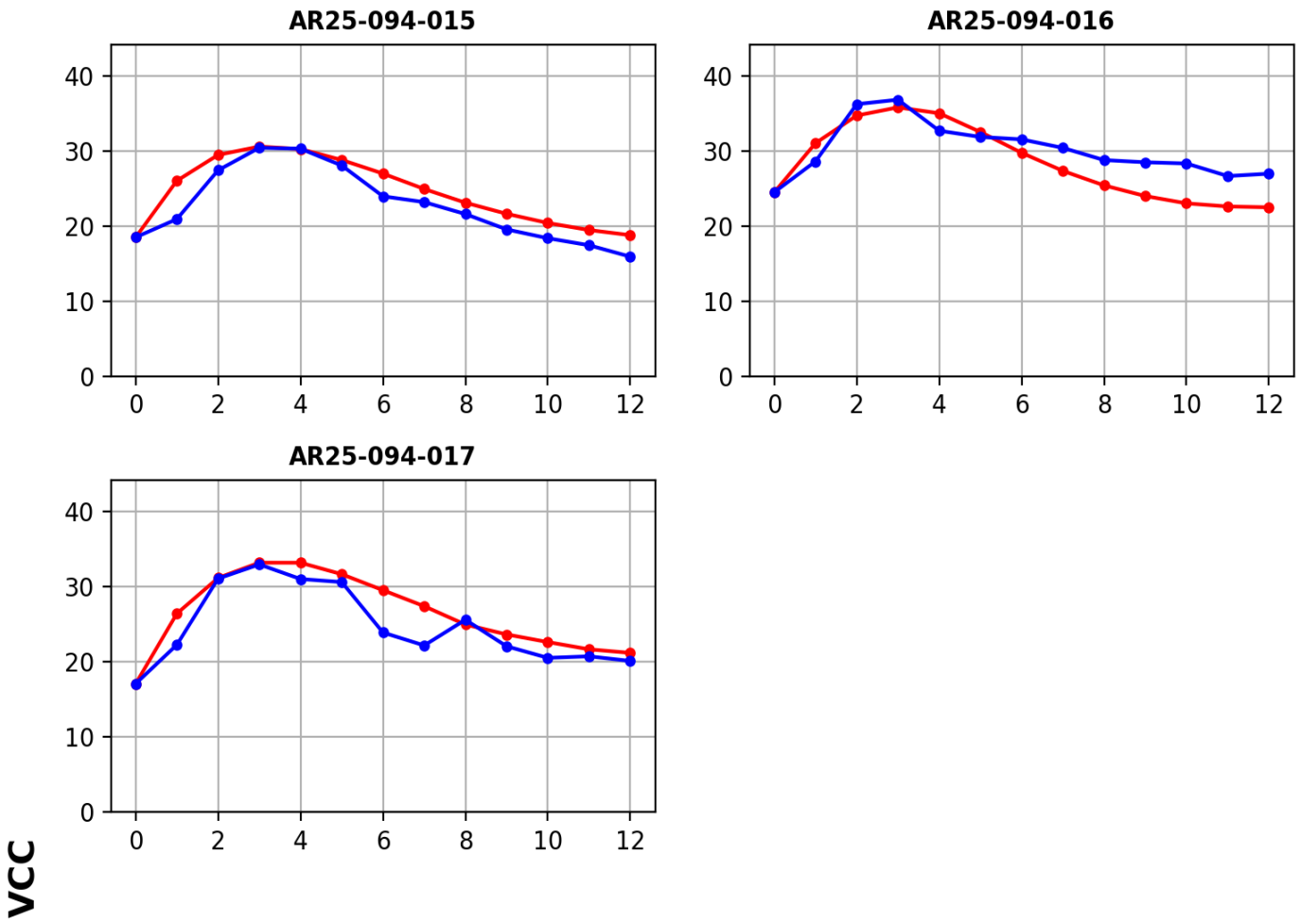

AR25-094-024
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
Lactate
Day

Model Testing Dataset

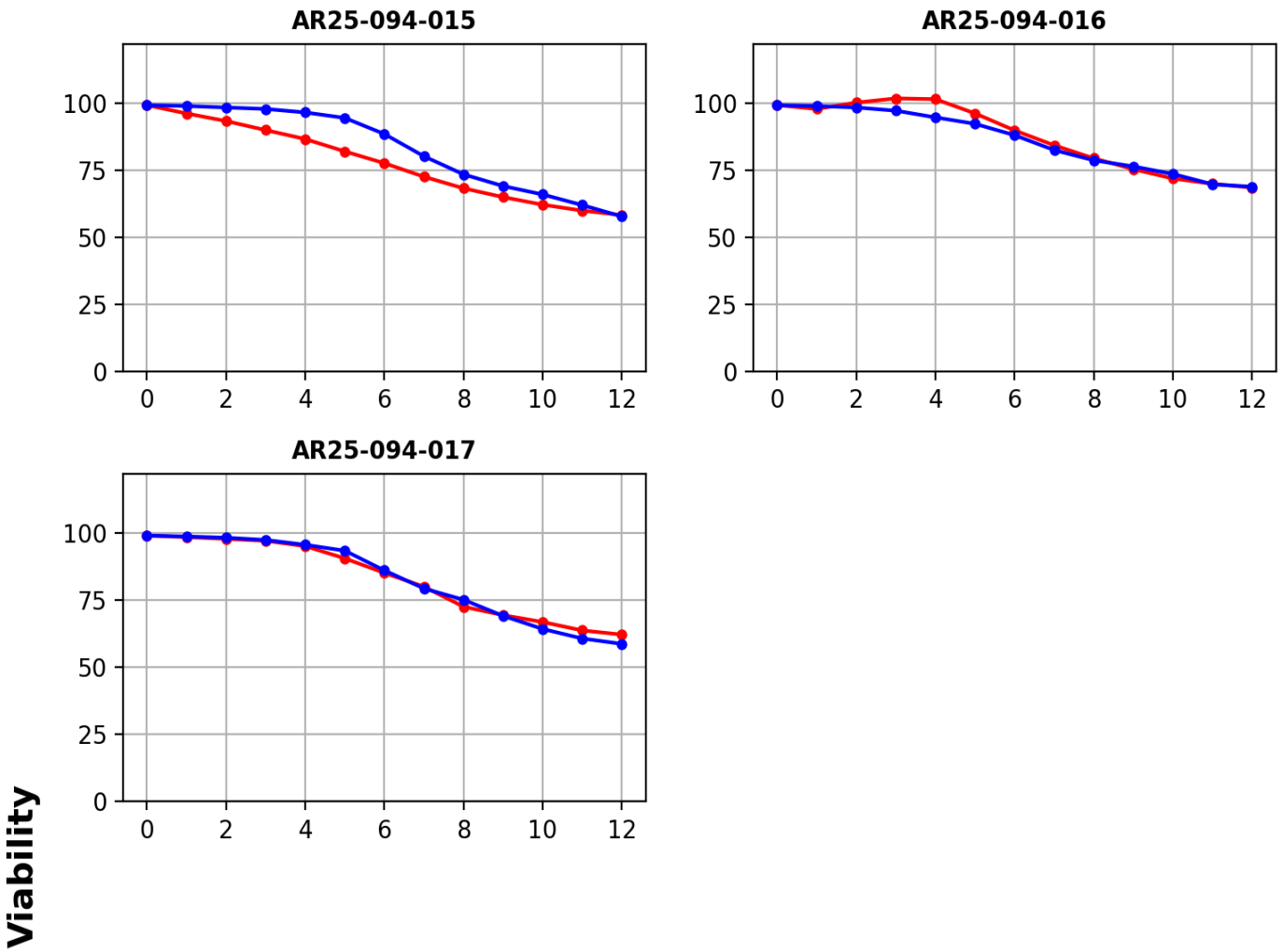
State: IGG


AR25-094-015
AR25-094-016
AR25-094-017
5000
4000
3000
2000
1000
0
0
2
4
6
8
10
12
IGG
Day

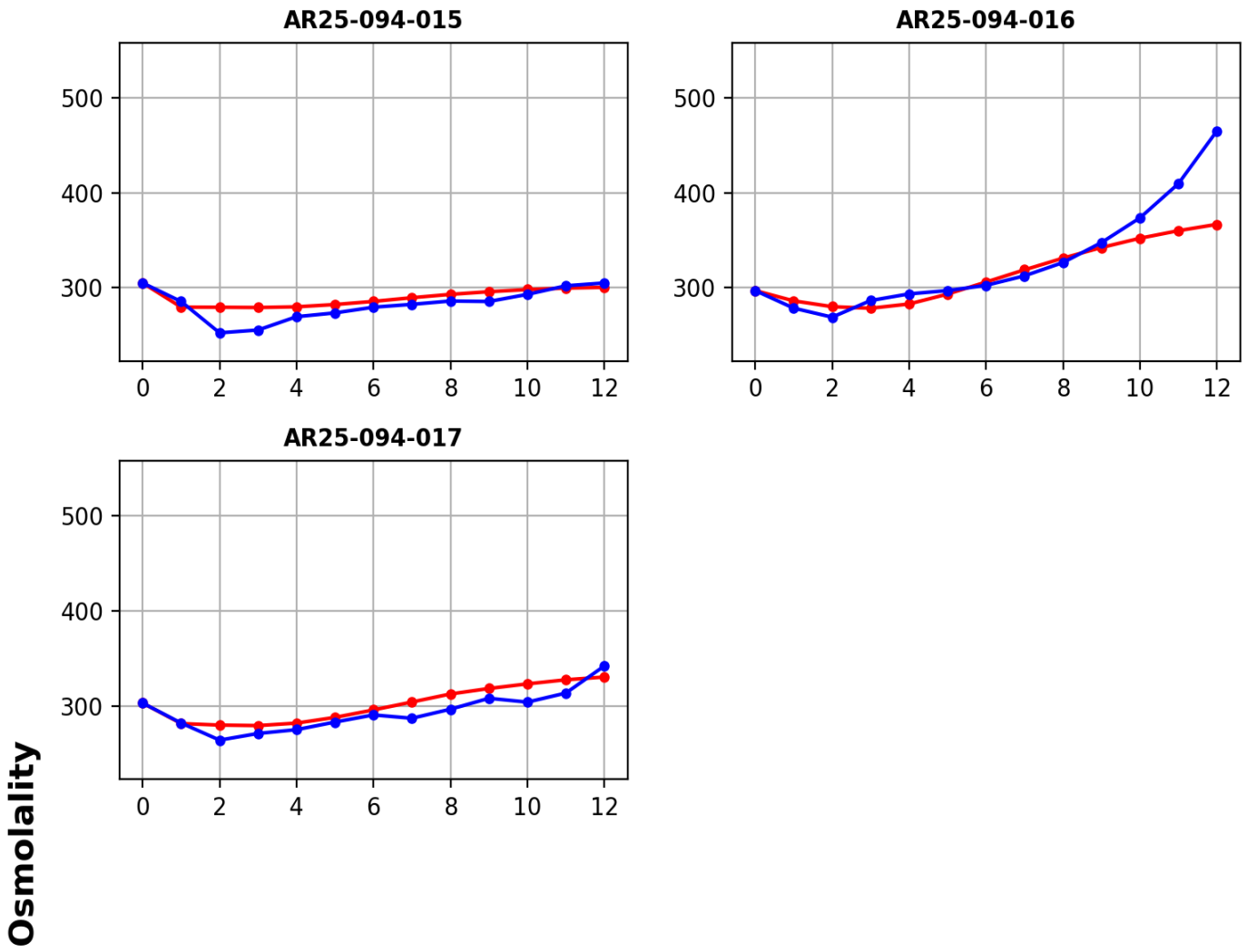
State: VCC


AR25-094-015
AR25-094-016
AR25-094-017
0
10
20
30
40
0
2
4
6
8
10
12
VCC
Day

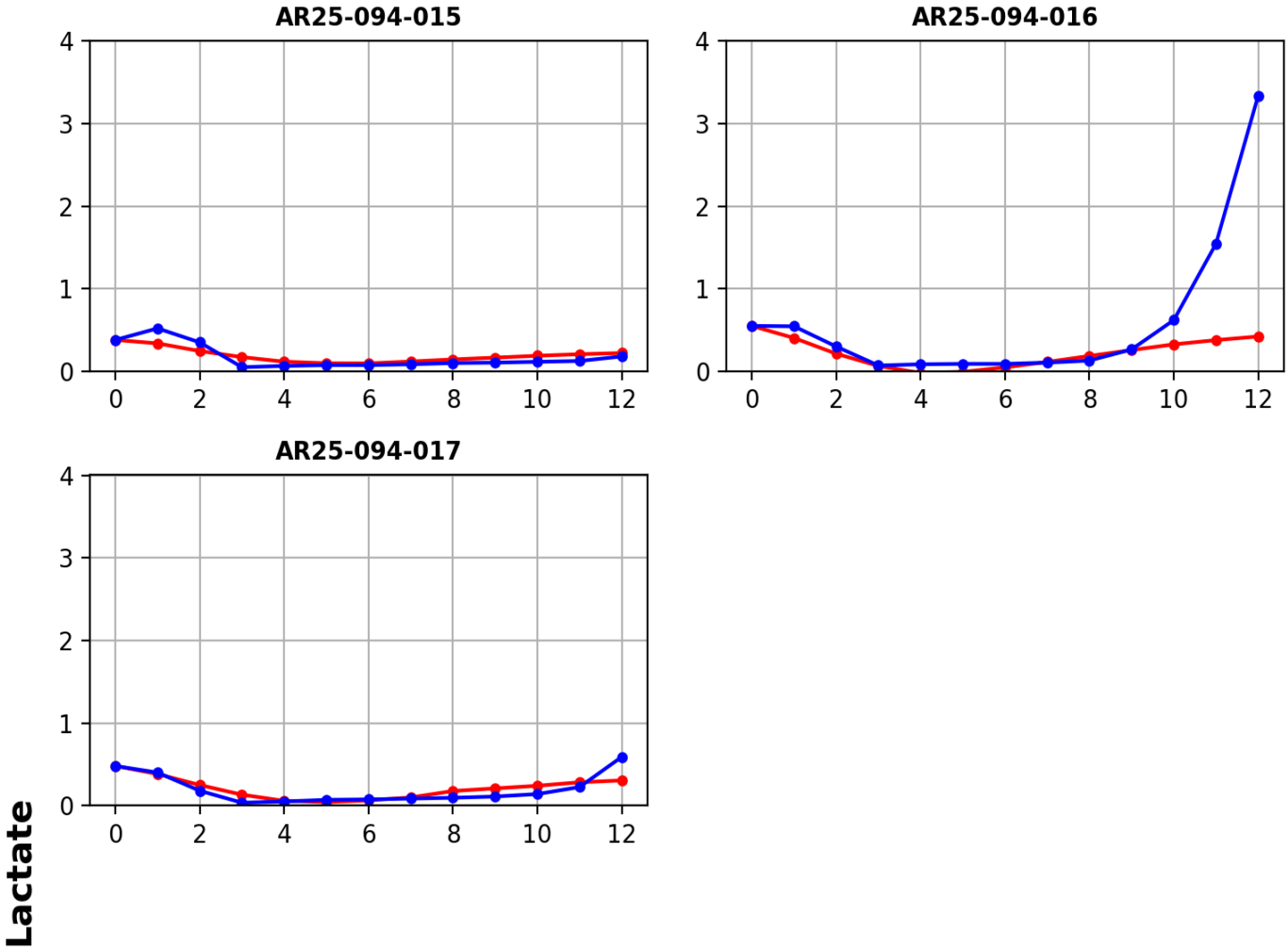
State: Viability



State: Osmolality



State: Lactate



Day

A Matrix

index	IGG	VCC	Viability	Osmolality	Lactate
IGG	1.78579	-0.14638	0.02383	-2.76816	1.30958
VCC	-1.63278	-0.31246	0.29817	2.07278	-0.44865
Viability	-1.01934	0.04628	-0.18269	0.89588	0.09832
Osmolality	6.70188	-0.1344	-0.33871	-10.0	4.57635
Lactate	-0.48283	-0.24393	0.21195	1.44422	-1.24601

B Matrix

index	Normalized_Daily_Feed	Temperature_setpoint	pH_setpoint
IGG	0.14984	0.36776	0.13115
VCC	0.13983	0.3864	-0.1708
Viability	0.44381	-0.25347	-0.1149
Osmolality	0.09926	-0.02442	0.54179
Lactate	-0.3211	-0.01504	-0.01399

Model RMSE Table

index	IGG	VCC	Viability	Osmolality	Lactate
AR25-094-013	171.31	2.79	2.9	18.68	0.1
AR25-094-014	148.88	0.92	1.54	11.53	0.13
AR25-094-015	327.81	2.25	6.74	11.77	0.08
AR25-094-016	318.31	3.08	2.75	31.68	0.88
AR25-094-017	123.1	2.65	1.85	11.68	0.1
AR25-094-018	88.05	1.81	3.68	8.69	0.06
AR25-094-019	118.92	3.06	2.29	11.71	0.18
AR25-094-020	221.49	2.45	1.99	10.1	0.09
AR25-094-021	190.85	3.1	3.53	7.98	0.08
AR25-094-022	181.58	3.21	6.6	18.08	0.28
AR25-094-023	201.42	2.31	3.86	12.5	0.09
AR25-094-024	150.16	3.2	1.83	11.81	0.07

Model R2 Table

index	IGG	VCC	Viability	Osmolality	Lactate
AR25-094-013	0.98	0.51	0.93	0.62	0.58
AR25-094-014	0.98	0.96	0.99	0.68	0.57
AR25-094-015	0.72	0.78	0.8	0.47	0.69
AR25-094-016	0.96	0.21	0.94	0.67	0.01
AR25-094-017	0.99	0.71	0.98	0.66	0.68
AR25-094-018	0.99	0.89	0.96	0.76	0.75
AR25-094-019	0.99	0.81	0.98	0.77	0.52
AR25-094-020	0.98	0.74	0.97	0.87	0.22
AR25-094-021	0.95	0.83	0.96	0.82	0.63
AR25-094-022	0.77	0.75	0.76	-0.8	0.03
AR25-094-023	0.98	0.85	0.88	0.88	0.82
AR25-094-024	0.98	0.76	0.98	0.43	0.79

Model Correlation Coefficient Table

index	IGG	VCC	Viability	Osmolality	Lactate
AR25-094-013	0.99	0.7	0.99	0.87	0.58
AR25-094-014	0.99	0.97	0.99	0.84	0.59
AR25-094-015	0.98	0.92	0.94	0.77	0.79
AR25-094-016	1.0	0.76	0.97	0.85	0.33
AR25-094-017	0.99	0.85	0.99	0.82	0.69
AR25-094-018	1.0	0.94	0.97	0.88	0.82
AR25-094-019	0.99	0.9	0.99	0.87	0.57
AR25-094-020	0.99	0.81	0.99	0.94	0.72
AR25-094-021	0.98	0.9	0.98	0.89	0.75
AR25-094-022	0.81	0.81	0.85	0.61	0.17
AR25-094-023	0.99	0.87	0.96	0.92	0.84
AR25-094-024	0.98	0.82	0.98	0.91	0.87